



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G1 SYSTEMS INTEGRATION AND ARCHITECTURE 1

FUNCTIONAL REQUIREMENTS SPECIFICATION (FRS)

Project Title: User Registration & Authentication

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1. Introduction

1.1. Purpose

The purpose of the **SpendWise** system is to provide college students and budget-conscious individuals with a secure and intuitive platform to manage personal finances. This document outlines the system architecture and user flows for registration, authentication, and expense tracking for developers and stakeholders.

1.2. Scope

SpendWise will allow users to register accounts, securely log in, and maintain a real-time dashboard of their spending. The system boundary includes user authentication and transaction management, while external payment gateways or bank integrations are currently out of scope.

1.3. Definitions, Acronyms, and Abbreviations

- **ERD**: Entity Relationship Diagram.
- **Auth**: Authentication/Authorization.
- **JWT**: JSON Web Token (used for secure session management).
- **RBAC**: Role-Based Access Control.

2. Overall Description

2.1. System Perspective

SpendWise operates as a standalone web-based financial management tool. It relies on a relational database to ensure data integrity for all financial transactions and user profiles.

2.2. User Classes and Characteristics

- **Guest User**: Unauthenticated individuals who can view the landing page and register for an account.
- **Authenticated User**: Registered users who have full access to the dashboard, expense logging, and history management.

2.3. Operating Environment

The system is designed to run on modern web browsers (Chrome, Firefox, Safari) and requires a backend server with a PostgreSQL or similar SQL database.

2.4. Assumptions and Dependencies

- Users have a stable internet connection.
- The system assumes all inputs are in a single base currency.

3. System Features and Functional Requirements

Describe each major feature of the system and its functional requirements.

3.1. Feature 1: User Management

Description: Provides secure access control through registration, login, and session termination.

Functional Requirements:

- The system allow users to register with a unique email and password
- The system shall verify credentials during login
- The system shall provide a logout function that destroys the active session

3.2. Feature 2: Transactional Logging (Add Expense)

Description: Enables users to record daily spending to maintain an accurate financial summary.

Functional Requirements:

- The system shall allow users to input an amount, category, and date for each expense.
- The system shall automatically update the “Total Spent” summary on the Dashboard upon successful logging.
- The system shall restrict access to this feature to Authenticated Users only.

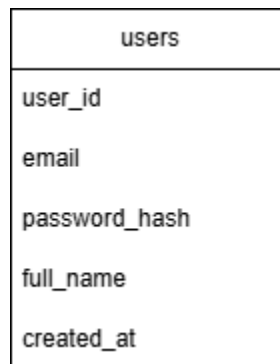
4. Non-Functional Requirements

- Security: All passwords must be hashed before storage; protected pages must redirect unauthenticated users to the login screen.
- Usability: The real-time dashboard summary should be clearly visible immediately upon login.
- Performance: Calculations for total balance and spent amounts should update in under 2 seconds.

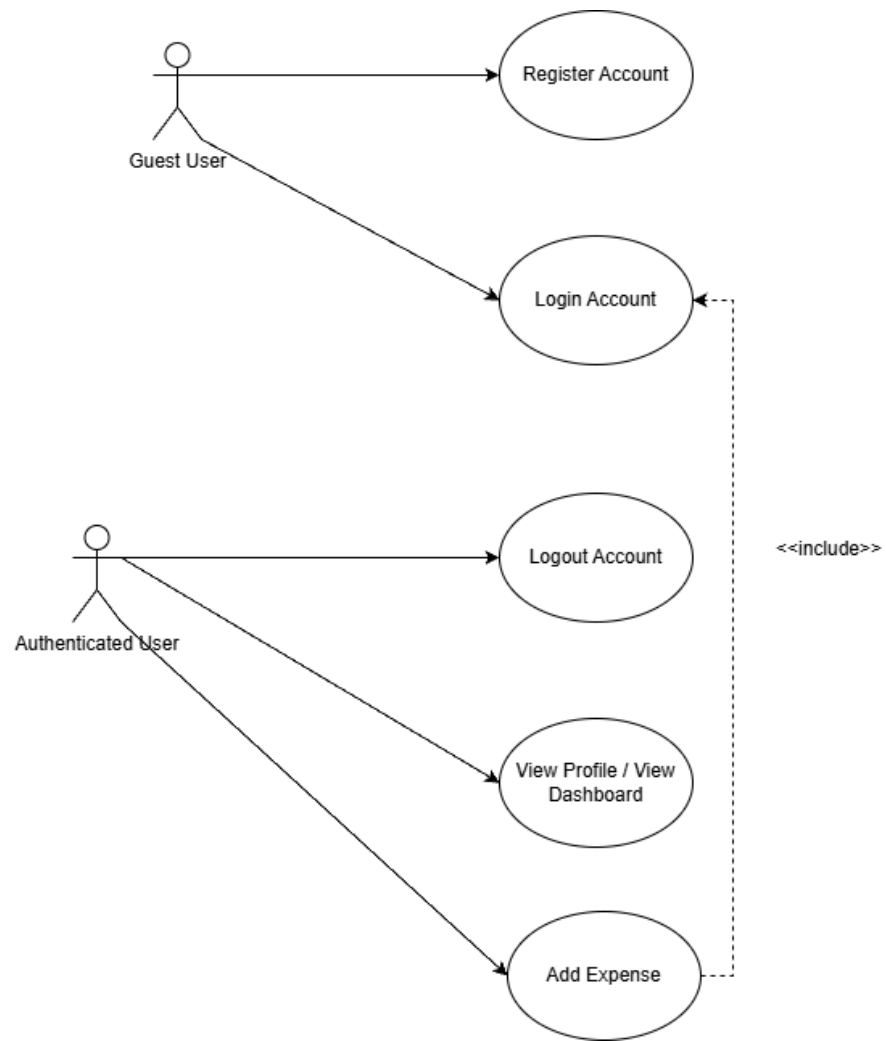
5. System Models (Diagrams)

Insert the necessary diagrams for the system:

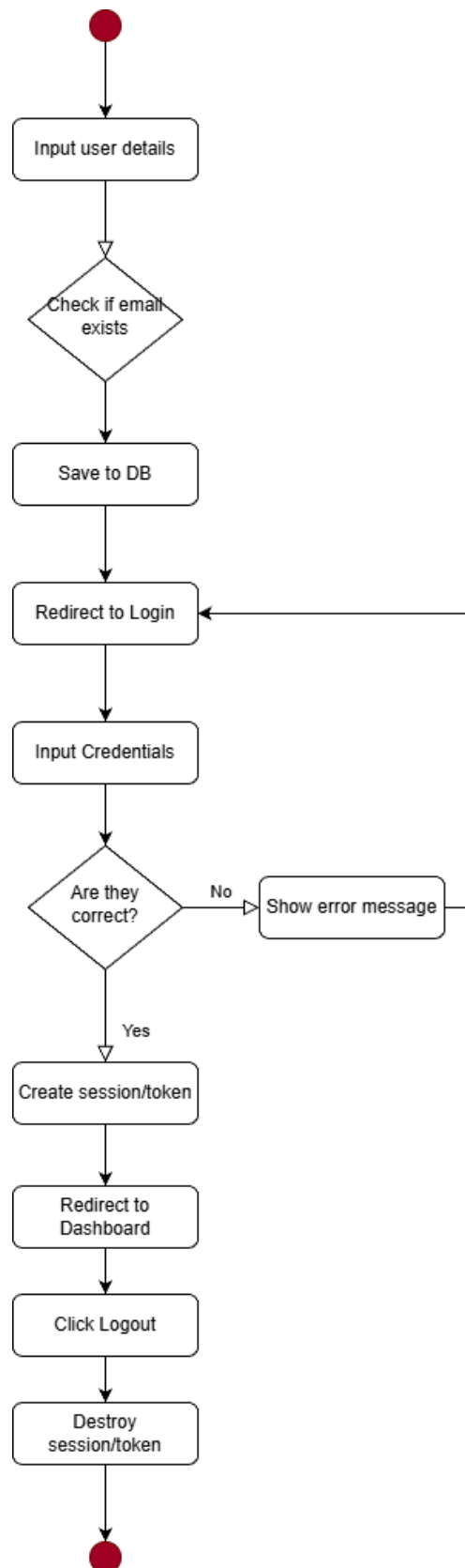
5.1. ERD



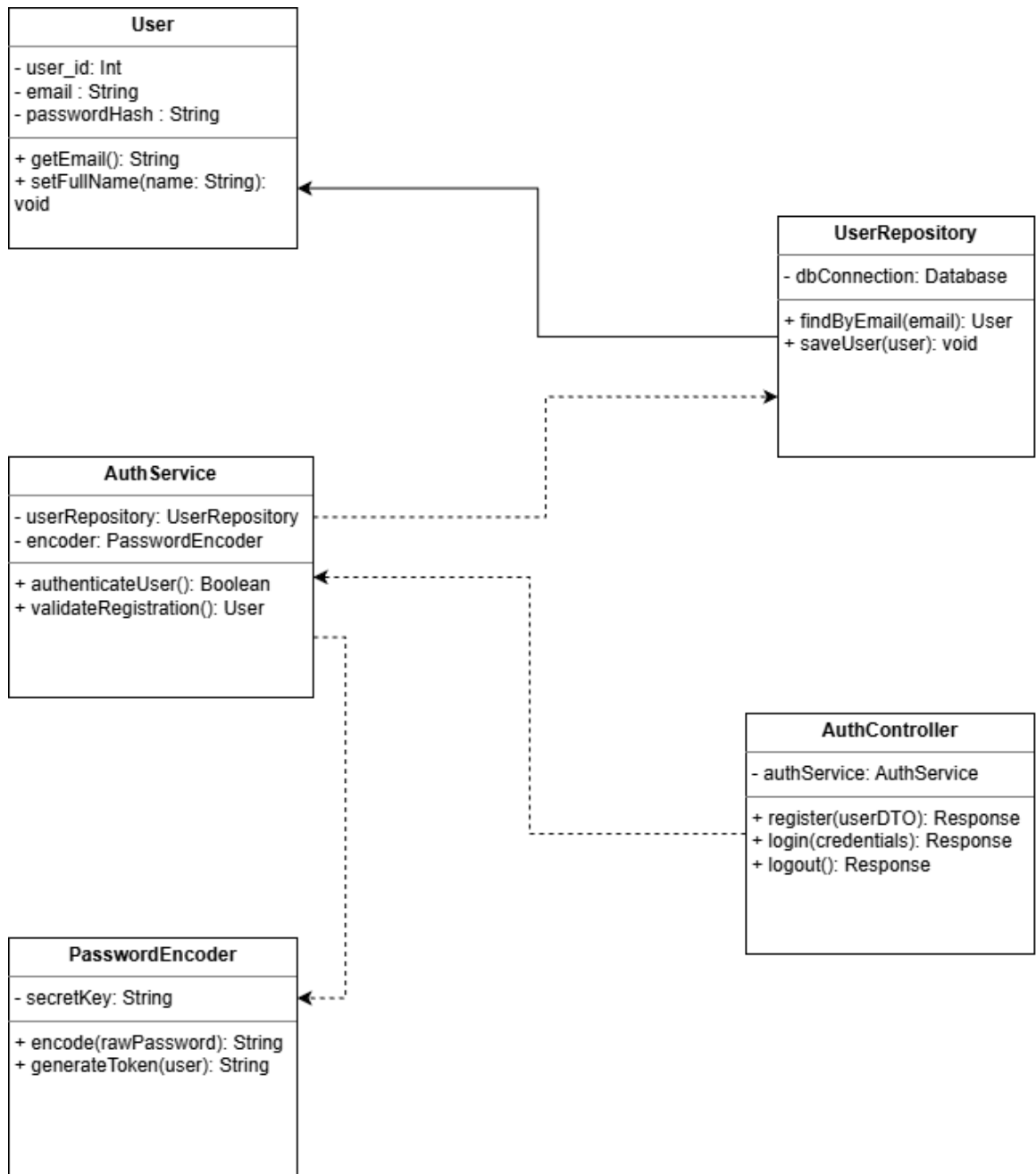
5.2. Use Case Diagram



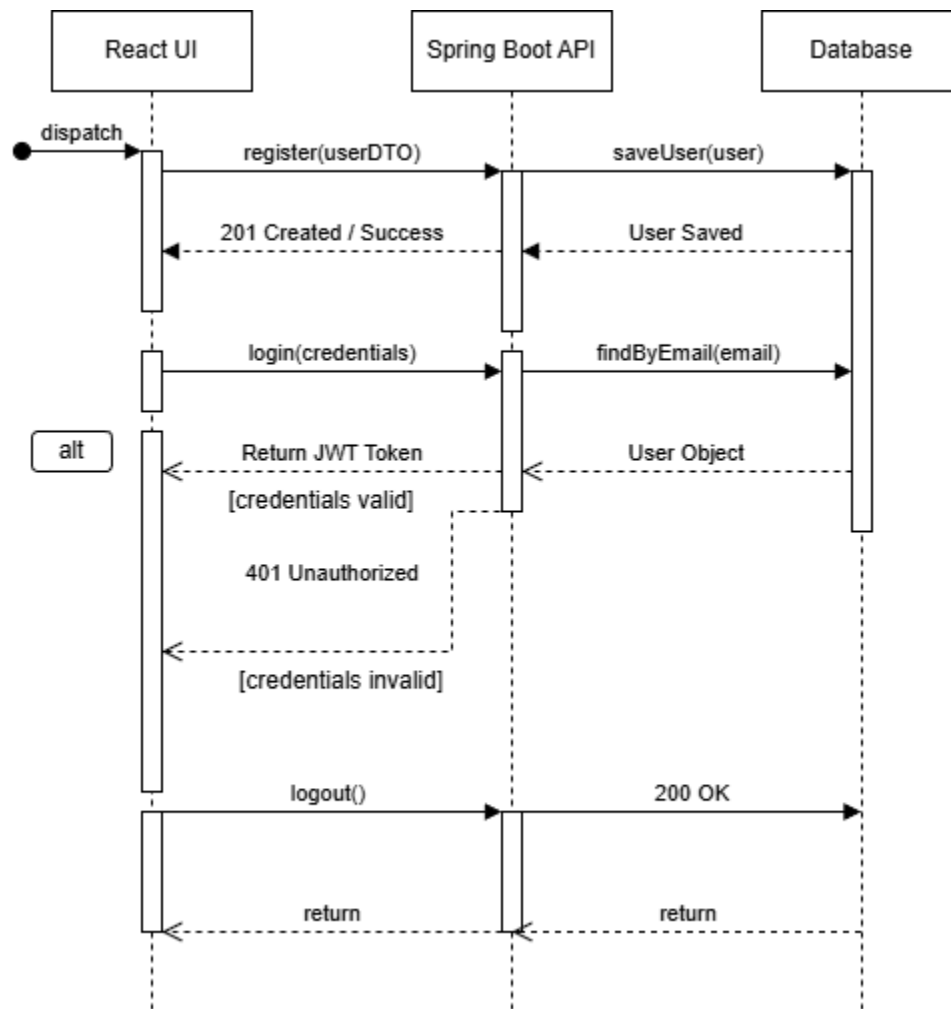
5.3. Activity Diagram



5.4. Class Diagram

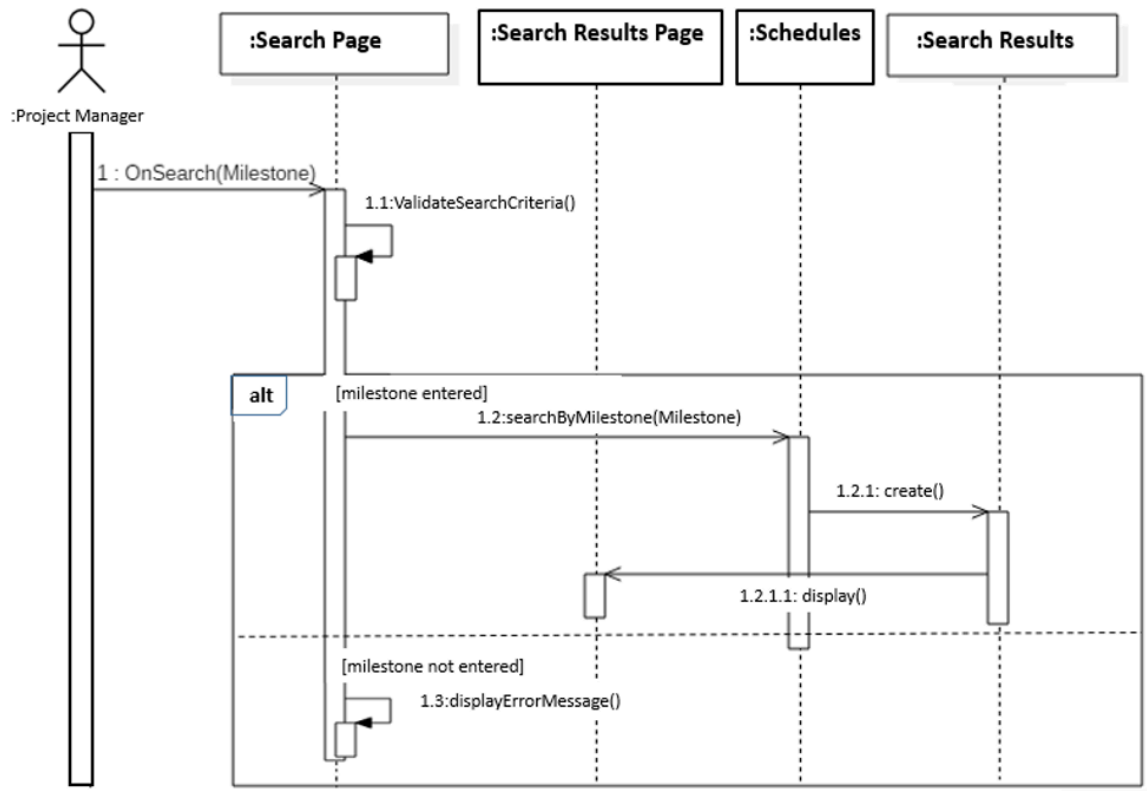


5.5. Sequence Diagram

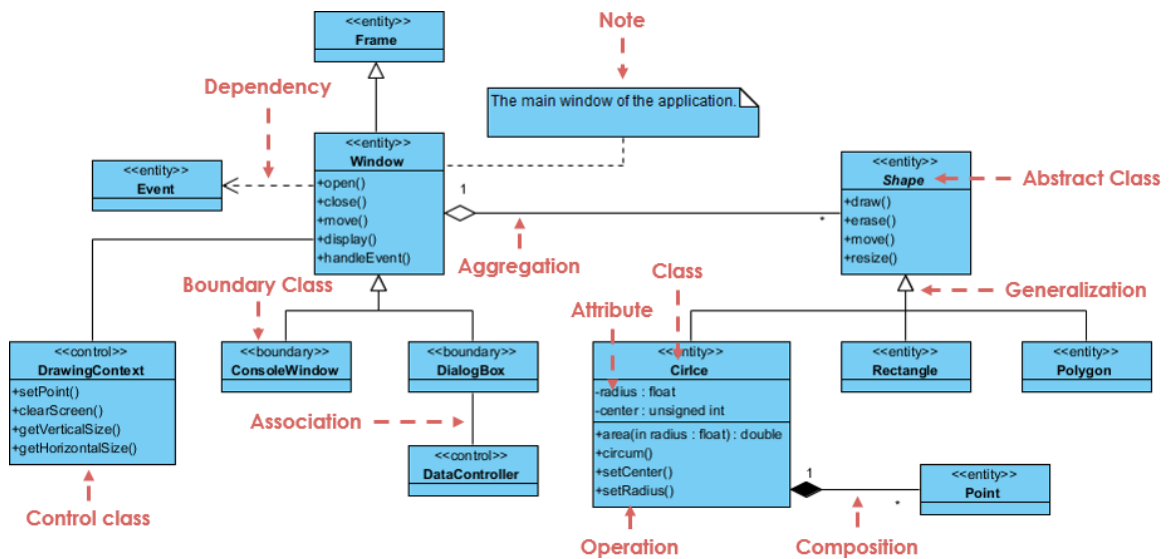


6. Appendices

6.1 Appendix A



6.2 Appendix B



6.3 Appendix C

